Lesson 1: Slope–Intercept Form and the Graph of a Line

Recognizing linear relations, reading m and b from a graph or an equation, and graphing a line from $y = mx + b$.

1. Recognizing Linear Relations

(Review from Unit 5.) A relation is **linear** when its graph is a straight line. Every non-vertical line can be written as $y = mx + b$ for constants $m, b \in R$; vertical lines have equations of the form $x = k$. Such equations are **linear equations**. In $y = mx + b$, x appears only to the first power, is never multiplied by another variable, never sits underneath a root or in a denominator, and never appears as an exponent. Every other relation is **non-linear**, and its graph is a curve rather than a straight line.

Example 1 — Sorting relations as linear or non-linear

Without graphing, sort each relation below into the LINEAR row or the NON-LINEAR row.

(a) $y = 5x - 3$ (b) $y = x^3$ (c) $y = 2x^2 + 7$ (d) $y = -\frac{2}{5}x + 6$ (e) $y = 9 - x$ (f) $y = \dfrac{3}{x+2}$ (g) $y = 7x$ (h) $y = 3^x$

LINEAR: _____

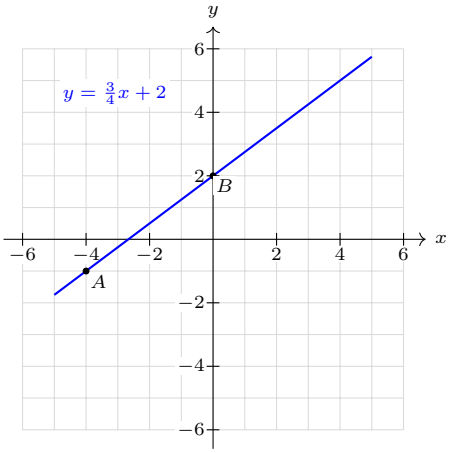
NON-LINEAR: _____

2. Reading Slope and y-intercept From a Graph

When a linear equation is already solved for y, use its graph to investigate what the numbers m and b in $y = mx + b$ tell you about the line. Examples 2 and 3 lead to the rule.

Example 2 — Slope and y-intercept from two marked points

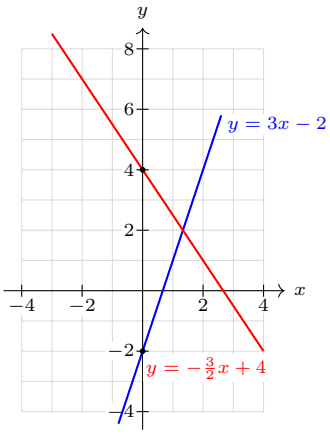
Naomi graphed the linear equation $y = \dfrac{3}{4}x + 2$, shown below with points A and B marked on the line.



(a) Use points A and B to determine the slope and y-intercept of the graph.

(b) Compare the values found in (a) with the coefficient of x and the constant term in $y = \dfrac{3}{4}x + 2$.

Example 3 — Two more lines, and a conjecture



(a) Two lines are graphed at the left. Use the grid to state the slope and y-intercept of each.

$y = 3x - 2$: slope = _____ y-intercept = _____

$y = -\frac{3}{2}x + 4$: slope = _____ y-intercept = _____

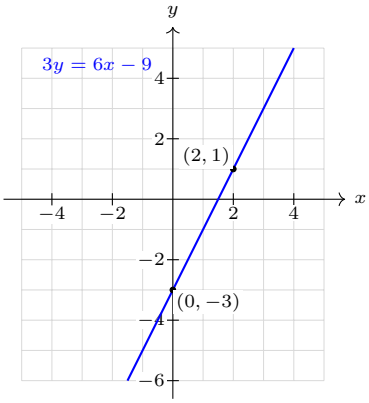
(b) Make a conjecture about the slope and y-intercept of the graph of the linear equation $y = mx + b$.

3. When the Equation Is Not Yet Solved For y

If a linear equation does not arrive in the form $y = mx + b$, the coefficient of x and the constant term you can see are *not* automatically the slope and y-intercept.

Example 4 — An equation that must be rearranged first

The graph of the linear equation $3y = 6x - 9$ is shown below.



(a) Use the graph to determine the slope and y-intercept of $3y = 6x - 9$.

(b) Explain why, in this case, the slope is not 6 (the coefficient of x) and the y-intercept is not -9 (the constant term).

Slope–Intercept Form of the Equation of a Line → $y = mx + b$

The graph of an equation in the form $y = mx + b$ (or of a function $f(x) = mx + b$) is a straight line with **slope** m and **y-intercept** b. This is the **slope–intercept form** of the equation of a line, and its graph can be drawn without a table of values. An equation that is not already solved for y must first be rearranged into this form before m and b can be read off directly.

Example 5 — Reading m and b from the equation alone

Determine the slope and y-intercept of the graph of each linear equation.

(a) $y = 4x + 5$

(b) $y = 9 - \frac{3}{5}x$

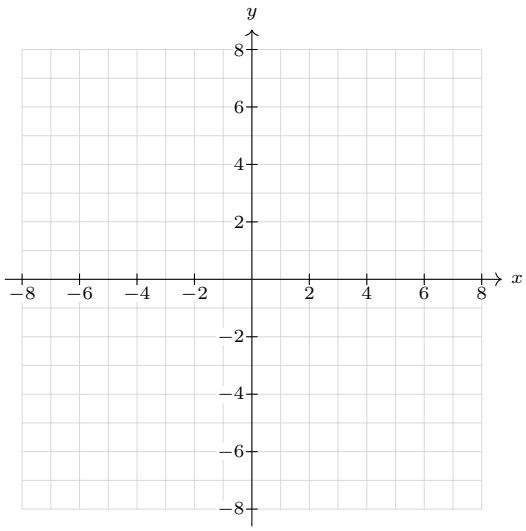
(c) $8y = 6x + 3$

4. Graphing a Line From Its Equation

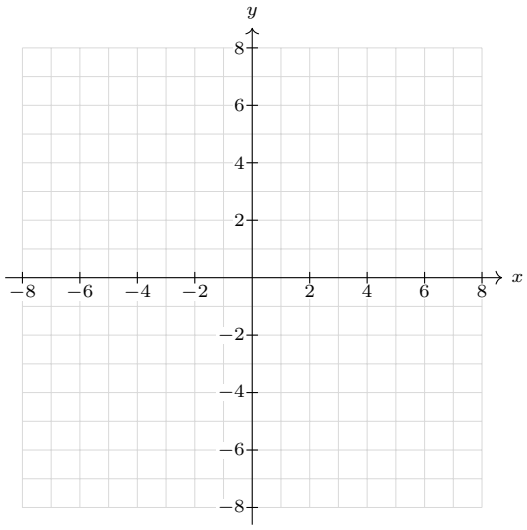
Once a line's equation is written as $y = mx + b$, it can be sketched without a table of values in either of two ways: plot the y-intercept $(0, b)$ and step out further points using the slope as $\frac{\text{rise}}{\text{run}}$; or find both the x-intercept and the y-intercept and join those two points. A negative slope such as $-\frac{3}{4} = \frac{-3}{4} = \frac{3}{-4}$ may be stepped off as down 3 and right 4, or as up 3 and left 4; both steps land on the same line.

Example 6 — Graphing from the slope, and from the two intercepts

(a) Consider $y = 3x - 4$. State the slope and y-intercept, mark the y-intercept on the grid, use the slope to mark three more points, and join them to graph the line.



(b) Consider $y = \frac{5}{6}x - 5$. State the y-intercept, determine the x-intercept algebraically, mark both intercepts on the grid, and join them to graph the line.



Lesson 2: From Slope to Equation: Building Linear Equations with $y = mx + b$

Building the equation of a line from its slope and intercept, from its graph, from a parallel or perpendicular condition, and for horizontal and vertical lines.

1. Building an Equation Directly from Slope and y-Intercept

When the slope m and the y-intercept b are both known, nothing is left to calculate: substitute both values into $y = mx + b$. A given point of the form $(0, k)$ *is* the y-intercept, so it supplies $b = k$ immediately.

Example 1 — Slope and y-intercept given directly

(a) Write the equation of the line that passes through $(0, -5)$ and has slope $\frac{3}{4}$.

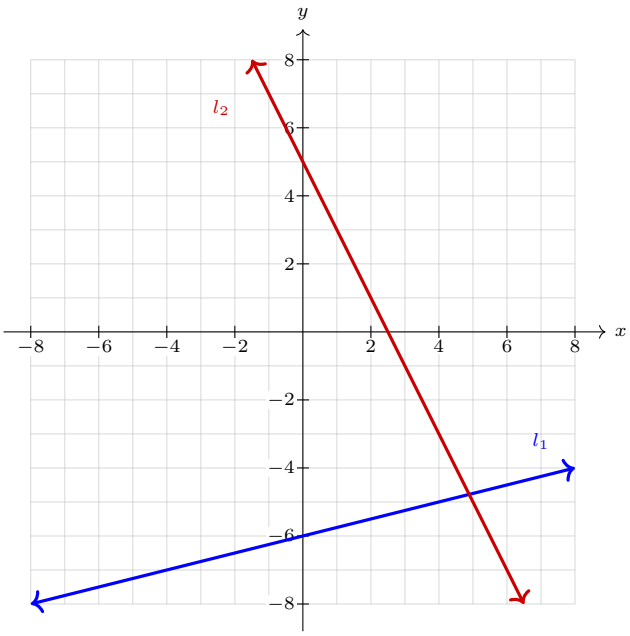
(b) Write the equation of the line with y-intercept 7 and slope $-\frac{2}{5}$.

2. Reading Slope and y-Intercept from a Graph

When a line passes through lattice points (points whose coordinates are both integers), count the rise and run between two of them to get m, and read b where the line crosses the y-axis. Substituting both into $y = mx + b$ gives the equation.

Example 2 — Reading m and b off a grid

Each line below passes through points with integer coordinates. For each line, state the slope and the y-intercept, and determine the equation of the line.



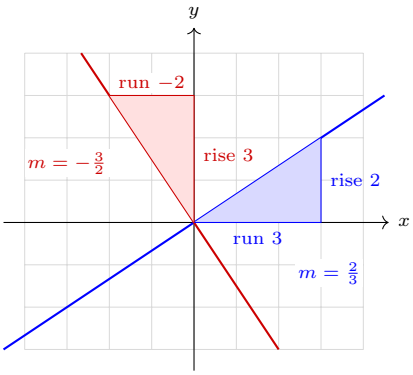
Line l_1:
$m =$ _____ $b =$ _____

Line l_2:
$m =$ _____ $b =$ _____

3. Parallel and Perpendicular Line Conditions

Two non-vertical lines are **parallel** exactly when their slopes are equal; all vertical lines are parallel. Two non-vertical lines are **perpendicular** exactly when their slopes are negative reciprocals of each other — that is, when their product is -1. A horizontal line and a vertical line are always perpendicular. Either condition supplies the slope of a new line from the equation, or from two points, of another line.

Why negative reciprocals? The blue line has slope $\frac{2}{3}$: run 3, rise 2. Turn its slope triangle 90° about the origin: the run of 3 becomes a *rise* of 3, and the rise of 2 becomes a *run* of 2 to the *left*. The turned (red) line is perpendicular to the blue one, and its slope is $\frac{3}{-2} = -\frac{3}{2}$. Rise and run swap places and one sign changes, so $\frac{2}{3} \times \left(-\frac{3}{2}\right) = -1$.



Example 3 — Slope from a parallel or perpendicular condition

Determine the equation of each line.

(a) the line parallel to $y = \frac{1}{2}x + 7$, with the same y-intercept as $y = 5x - 3$

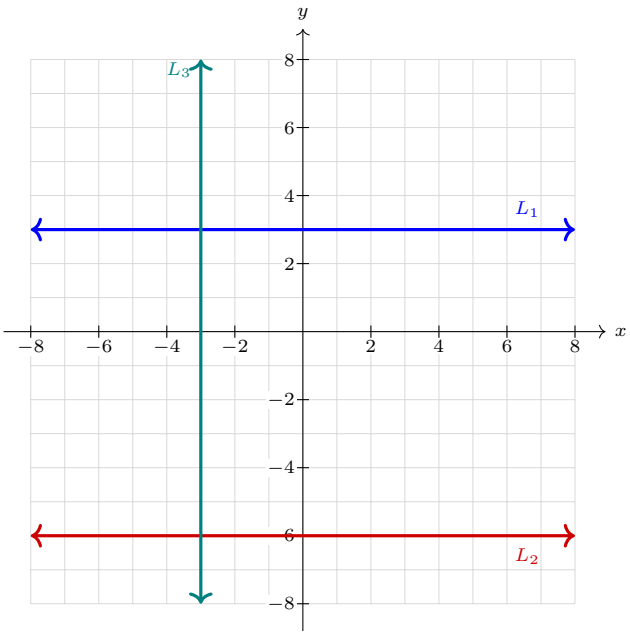
(b) the line passing through $(0, 11)$, perpendicular to the line joining $(3, -4)$ and $(-6, 2)$

4. Horizontal and Vertical Lines

Horizontal and vertical lines are special cases. Use the grid in Example 4 to work out what their slopes are, and what their equations look like.

Example 4 — Horizontal and vertical lines

The grid shows two horizontal lines, L_1 and L_2, and one vertical line, L_3.



(a) State the slope and y-intercept of the horizontal line L_1.

(b) Use $y = mx + b$ to determine the equation of L_1.

(c) Predict the equation of the horizontal line L_2.

(d) State the slope and y-intercept of the vertical line L_3.

(e) Explain why the form $y = mx + b$ cannot be used to determine the equation of L_3.

(f) Predict the equation of the vertical line L_3.

Summary — horizontal and vertical lines. A horizontal line rises 0 units for every unit of run, so $m = 0$ and $y = mx + b$ collapses to $y = b$. Every horizontal line has an equation of the form $y = k$, and it crosses the y-axis at $(0, k)$. Any two points of a vertical line have a run of 0, and division by 0 is undefined, so the slope of a vertical line is **undefined**. With no value of m to substitute, it cannot be written in the form $y = mx + b$; instead every vertical line has an equation of the form $x = k$, and it crosses the x-axis at $(k, 0)$.

Lesson 3: The General Form of a Linear Equation

Writing $Ax + By + C = 0$, reading slope and intercepts back out of it, and using it to test parallel and perpendicular lines.

1. Rewriting an Equation in General Form

A linear equation is in **general form** once every term has been gathered onto one side, leaving the other side equal to zero:

$$Ax + By + C = 0.$$

The conventions for general form are:

- A, B and C are **integers** (clear any fractions first);
- $A \geq 0$ — the x-term is positive. A horizontal line has $A = 0$; then make B positive instead: write $y - 4 = 0$, not $-y + 4 = 0$;
- A, B and C have **no common factor** other than 1 — divide any common factor out. For example, $10x - 4y + 20 = 0$ is written $5x - 2y + 10 = 0$.

Unlike $y = mx + b$, general form can represent *every* line in the plane, horizontal and vertical lines included.

General Form of the Equation of a Line $\rightarrow$ $Ax + By + C = 0$

To convert from slope–intercept form to general form, move every term to the left side. If the slope or y-intercept is a fraction, first multiply the entire equation by the common denominator so that A, B and C all come out as integers. If A comes out negative, multiply every term by -1 (or move the terms to the right side instead).

Example 1 — Slope–intercept form to general form

Convert each equation from slope–intercept form, $y = mx + b$, to general form $Ax + By + C = 0$, where A, B and C are integers with no common factor.

(a) $y = 4x - 9$ **(b)** $y = \frac{3}{5}x + 2$ **(c)** $y = -\frac{1}{3}x + \frac{5}{6}$

(d) Jordan converted $y = 2x - 6$ to general form and wrote $4x - 2y - 12 = 0$. Is Jordan's equation the same line? Is it written in general form? Explain, and correct it if needed.

2. Extracting Slope and Intercepts from General Form

Once an equation is in general form the slope and y-intercept are recovered by solving for y — exactly the reverse of Example 1. Isolate the By term, then divide every term by B; the result is in slope–intercept form. Either intercept can also be read straight off the general form without solving for y: substitute $y = 0$ for the x-intercept, and $x = 0$ for the y-intercept.

Example 2 — Slope and y-intercept from general form

(a) & **(b)** Determine the slope and y-intercept of the graph of each line.

(a) $3x - 4y + 20 = 0$ **(b)** $8x + 3y - 21 = 0$

(c) The lines $4x - 5y + 15 = 0$ and $6x - ky - 9 = 0$ have the same y-intercept. Determine the value of k.

General rule. For the line $Ax + By + C = 0$ (with $B \neq 0$), solving for y gives

$$By = -Ax - C \quad \Longrightarrow \quad y = -\frac{A}{B}x - \frac{C}{B}, \qquad \text{so} \qquad m = -\frac{A}{B}, \qquad y\text{-intercept} = -\frac{C}{B}.$$

Setting $y = 0$ gives $Ax + C = 0$, so the x-intercept is $-\dfrac{C}{A}$ (when $A \neq 0$). Use the rule as a quick check on your rearranging: in Example 2(a), $3x - 4y + 20 = 0$ gives $m = -\frac{3}{-4} = \frac{3}{4}$ and y-intercept $-\frac{20}{-4} = 5$.

3. Perpendicular Lines in General Form

Two non-vertical lines are **parallel** exactly when their slopes are equal; all vertical lines are parallel. Two non-vertical lines are **perpendicular** exactly when their slopes are negative reciprocals of one another, that is, when their slopes multiply to give -1; a horizontal line and a vertical line are always perpendicular. Since the slope can be pulled out of any general-form equation, lines given in general form are tested by comparing those slopes.

Example 3 — Testing three lines for perpendicularity

Which of the following lines is/are perpendicular to the line $5x - 3y + 6 = 0$?

(i) $3x + 5y - 8 = 0$ **(ii)** $x + 3y - 9 = 0$ **(iii)** $2x - 5y + 10 = 0$

4. Writing an Equation from Given Conditions

A line whose slope is described only indirectly — through a parallel or perpendicular relationship to another line — and whose y-intercept must match that of a third line is built in four steps: determine the target slope, determine the target y-intercept, assemble the equation in slope–intercept form, then convert to general form if required.

Example 4 — Building a line from a perpendicular and an intercept

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Write the equation of a line which is perpendicular to $3x + 4y - 8 = 0$ and has the same y-intercept as $5x + y - 10 = 0$. Answer in slope–intercept form and in general form.

General form is also useful for quickly checking claims about a line — its slope, its intercepts, or whether it passes through a given point — without first solving for y.

Example 5 — Judging three statements about one line

A student made the following statements about the line with equation $4x - 3y + 18 = 0$.
Statement 1: The line has a slope of 4.
Statement 2: The line is parallel to $8x - 6y + 15 = 0$.
Statement 3: The line passes through $(3, 10)$.

Which of the above statement(s) is/are true?
A. 1 and 2 only **B.** 1 and 3 only **C.** 2 and 3 only **D.** some other combination

Lesson 4: Using Slope–Point Form to Determine the Equation of a Line

Building $y - y_1 = m(x - x_1)$ from the slope formula — then using it to convert indirect descriptions of a line into general form.

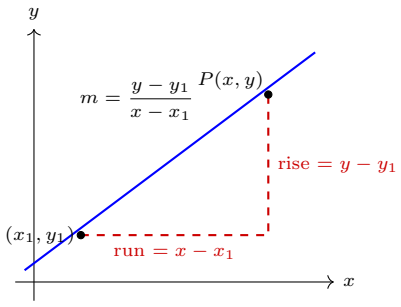
1. Building the Equation from the Slope Formula

Let (x_1, y_1) be a fixed point on a line and let $P(x, y)$ be any other point on that same line, free to move along it. From (x_1, y_1) to P the rise is $y - y_1$ and the run is $x - x_1$, so the slope m between them is

$$m = \frac{y - y_1}{x - x_1}.$$

Multiplying both sides by $(x - x_1)$ clears the fraction and produces an equation that is also satisfied by (x_1, y_1) itself (both sides become 0), so it is the equation of the entire line:

$$y - y_1 = m(x - x_1).$$



Walkthrough. A line has slope 4 and passes through $A(2, -1)$. With $P(x, y)$ any point on the line,

$$\frac{y - (-1)}{x - 2} = 4 \quad \Longrightarrow \quad y + 1 = 4(x - 2) \quad \Longrightarrow \quad y + 1 = 4x - 8 \quad \Longrightarrow \quad y = 4x - 9.$$

The middle equation, $y+1 = 4(x-2)$, is exactly the pattern $y-y_1 = m(x-x_1)$ with $m = 4$ and $(x_1, y_1) = (2, -1)$ substituted in directly — no cross-multiplying or solving required.

Slope–Point Form (also called *point-slope form*). $y - y_1 = m(x - x_1)$, where m is the slope and (x_1, y_1) is *any* known point on the line.

- Use it whenever a slope and a point are given — it requires no solving, only substitution.
- A final answer is usually expected in general form, $Ax+By+C = 0$, or in slope–intercept form, $y = mx+b$, so slope–point form is often just the first step.
- Any point on the line may be used for (x_1, y_1); different choices produce different-*looking* slope–point equations that all simplify to the same line.

2. Practice with the Slope–Point Form

Example 1 — Writing slope–point form from a point and a slope

State the equation, in slope–point form, of the line through the given point with the given slope.

(a) $(4, 7)$, slope 5 **(b)** $(2, -3)$, slope -6 **(c)** $(-8, -2)$, slope $\frac{1}{3}$

Example 2 — Converting a point and a slope to $y = mx + b$

In each case the slope of a line and a point on the line are given. Determine the equation of the line in slope–intercept form, $y = mx + b$.

(a) $m = 6$, point $(-4, 3)$ **(b)** $m = -8$, point $(-1, 5)$

Example 3 — Two methods compared: $y = mx + b$ against $y - y_1 = m(x - x_1)$

Priya and Deshawn were each asked: *"Determine the equation, in general form, of the line with slope -5 passing through the point $(4, -2)$."* Priya could only remember $y = mx + b$; Deshawn remembered $y - y_1 = m(x - x_1)$. Complete each student's method, then compare.

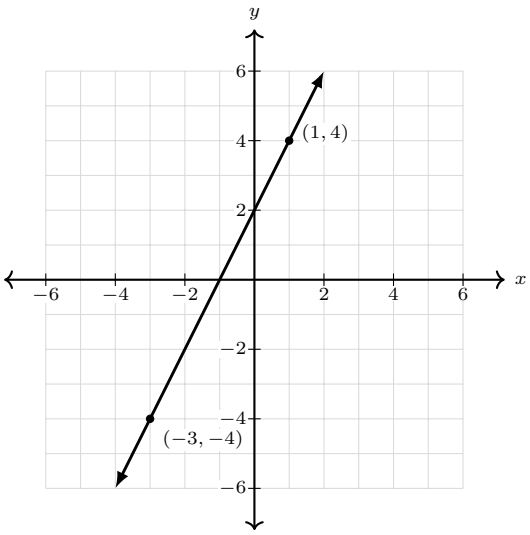
(a) Priya's method. Start from $y = mx + b$, substitute the slope, then substitute the point to solve for b.

(b) Deshawn's method. Start from $y - y_1 = m(x - x_1)$ and substitute the slope and the point directly.

(c) Convert both final answers to general form, $Ax + By + C = 0$. What do you notice?

Example 4 — From a grid to general form; reading m and (x_1, y_1)

(a) The line on the grid passes through at least two points with integer coordinates. Determine the equation of the line in general form.



(b) Each equation below is written in slope–point form. State the slope of the line and the coordinates of the point used to write the equation.

(i) $y - 5 = -\frac{2}{3}(x + 9)$ **(ii)** $y + 7 = 4x$ **(iii)** $y = -6(x - 10)$

3. Using Slope–Point Form for Indirect Descriptions of a Line

Three forms are now available:

$Ax + By + C = 0$ → General form of a linear equation.
$y = mx + b$ → Slope–intercept form of a linear equation.
$y - y_1 = m(x - x_1)$ → Slope–point form of a linear equation.

Slope–intercept form is the natural choice when the slope m and the y-intercept b are both handed to you. Slope–point form is more flexible: it needs the slope and any single point (x_1, y_1), which need not be the y-intercept.

Most questions do not state a slope and a point outright. Instead the line is described indirectly — through two points on it, through its relationship (parallel or perpendicular) to another line, or through an intercept it shares with another line. The work is turning those clues into a slope and a point; slope–point form then produces the equation, and clearing fractions and collecting terms converts it to general form.

Example 5 — General form of the line through two points

Given the points $P(4, -3)$ and $Q(-2, 9)$, determine the equation, in general form, of the line passing through P and Q.

Example 6 — General form from a slope and an x-intercept

Determine the equation, in general form, of the line with slope $\frac{3}{4}$ and an x-intercept of 8.

Example 7 — General form from a point and a perpendicular line

Determine the equation, in general form, of the line that passes through the point $(-4, 2)$ and is perpendicular to the line with equation $2x - 3y + 12 = 0$.

Example 8 — A perpendicular slope and a shared x-intercept

Determine the equation, in general form, of the line that is perpendicular to the line $4x - 9y - 27 = 0$ and has the same x-intercept as the line $3x - 5y - 15 = 0$.

Lesson 5: Graphing Linear Equations: A Strategy for Every Form

Choosing the fastest hand-graphing strategy for the form you are given — and recovering an equation from a table of data.

1. Three Ways to Write the Same Line

Every non-vertical line can be written in any of the three forms below, and each one advertises different information up front.

$Ax + By + C = 0$	General form — hides the slope and the intercepts, but both intercepts drop out by substitution.
$y = mx + b$	Slope–intercept form — hands you the slope m and the y-intercept b.
$y - y_1 = m(x - x_1)$	Slope–point form — hands you the slope m and one point (x_1, y_1) on the line, not necessarily an intercept.

The graphing strategy should exploit whichever piece of information the given form makes easiest to read, so that the equation never has to be rearranged at all.

2. Graphing by Hand Without Changing the Form

Choosing a Graphing Strategy from the Form

Given form	Strategy — graph it without rearranging
Slope–intercept $y = mx + b$	Plot the y-intercept $(0, b)$. From there step the slope $m = \frac{\text{rise}}{\text{run}}$ repeatedly to plot more points, then join them.
Slope–point $y - y_1 = m(x - x_1)$	Plot the given point (x_1, y_1). From there step the slope m repeatedly to plot more points, then join them.
General $Ax + By + C = 0$	Substitute $y = 0$ and solve for the x-intercept; substitute $x = 0$ and solve for the y-intercept. Plot both intercepts and draw the line through them. *If $C = 0$, both intercepts are $(0, 0)$; if an intercept is a fraction, it is hard to plot. Then substitute a convenient x to find another point with integer coordinates, or solve for y (Example 2).*
Horizontal or vertical $y = k$ or $x = k$	$y = k$ is horizontal through $(0, k)$, slope 0; $x = k$ is vertical through $(k, 0)$, slope undefined. Plot that single point and rule the line.

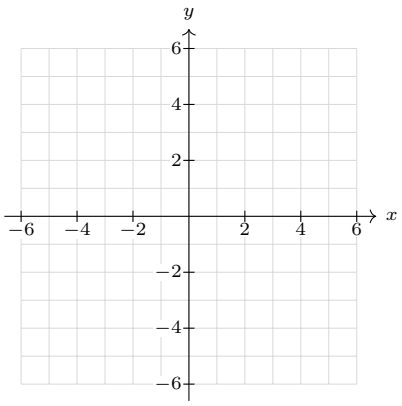
All of these strategies can describe the very same line. The three equations in Example 1 are one line written three ways, so

the three graphs must come out identical.

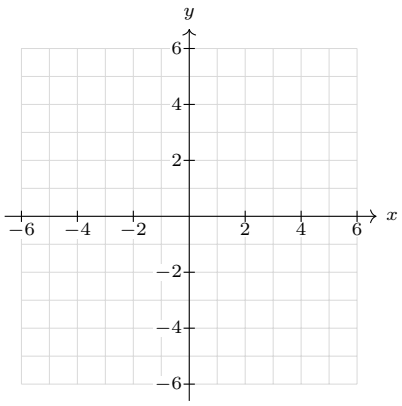
Example 1 — One line, three forms, three strategies

Without altering the form in which each equation is written, state the strategy you would use to graph it *by hand*, then draw the graph on the grid provided.

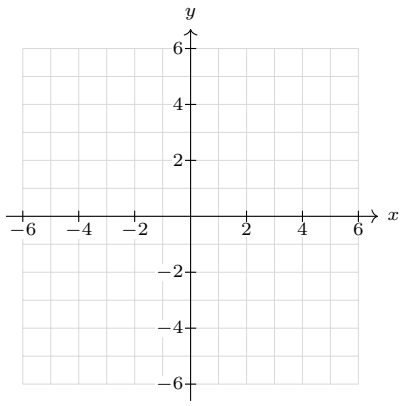
(a) $y = 3x - 6$



(b) $y + 3 = 3(x - 1)$



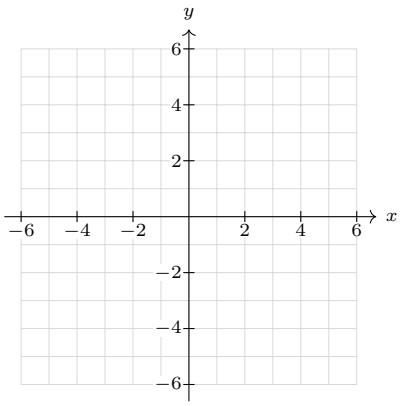
(c) $3x - y - 6 = 0$



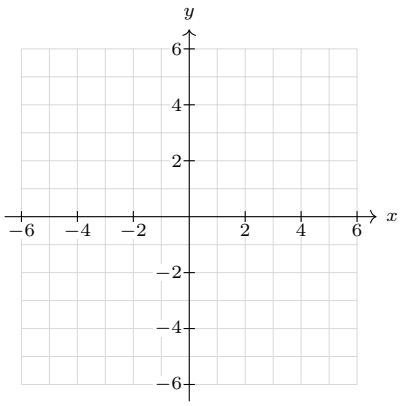
Example 2 — When the quick strategy needs care

Graph each line by hand, without rearranging it. In (a), watch the signs; in (b) and (c), the intercepts alone are not enough — find another point.

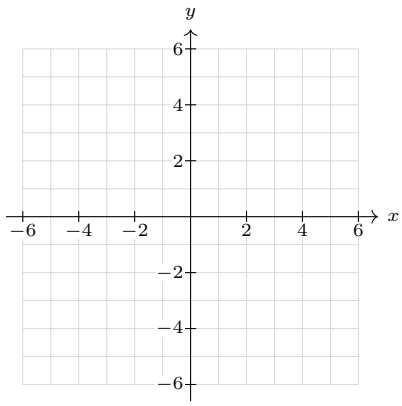
(a) $y + 1 = -\frac{2}{3}(x - 4)$



(b) $3x - 4y = 0$



(c) $2x + 3y - 4 = 0$



3. Reading a Graph Straight From Its Equation

Slope and y-intercept alone are enough to picture a line: the sign of m says whether it rises or falls, and the size of m says how steep it is. Rewriting each equation as $y = mx + b$ is therefore enough to match equations to graphs without plotting a single point.

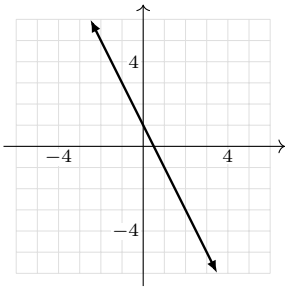
Example 3 — Matching equations to graphs

Match each linear relation to its graph below. Each graph is used exactly once.

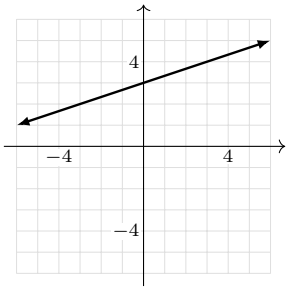
Equation 1: $3x - y + 2 = 0$ **Equation 2:** $y = -2x + 1$

Equation 3: $y - 1 = -\frac{1}{3}(x - 3)$ **Equation 4:** $3y - x - 9 = 0$

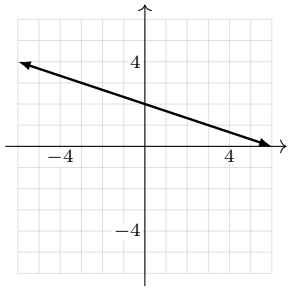
Graph A



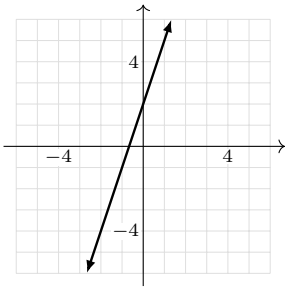
Graph B



Graph C



Graph D



Eqn 1 → Graph _____ Eqn 2 → Graph _____ Eqn 3 → Graph _____ Eqn 4 → Graph _____

Show the slope and y-intercept you used for each equation.

4. From a Table of Data to an Equation, and Back Again

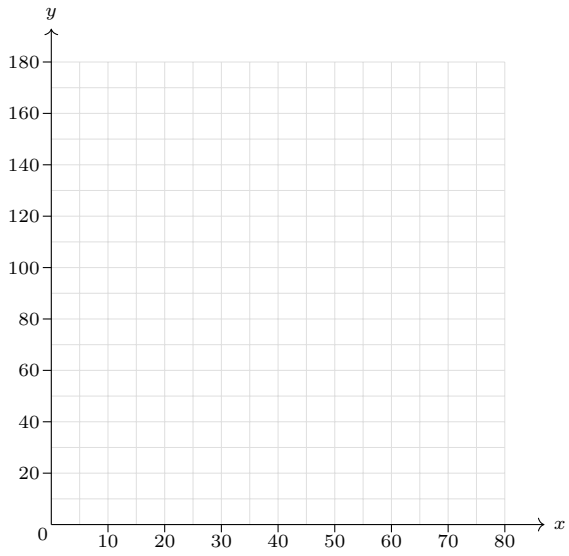
A linear relation can also be recovered from a table of values: plot the ordered pairs, confirm they are collinear, take any two of them to find the slope, and build the equation in whichever form is convenient before converting to general form. Solving that finished equation for the other variable reverses the relationship.

Example 4 — From data to an equation, and back

The table records values of x and y taken from a continuous linear relationship.

x	0	10	20	30	40	50	60	70
y	32	50	68	86	104	122	140	158

(a) Plot the data on the grid and determine, in general form, the equation of the linear relation the data represents.



(b) Rewrite the equation of the linear relation (i) in terms of x, that is $y = \ldots$, and (ii) in terms of y, that is $x = \ldots$.

(c) The formulas in (b) are well known in science. What quantities do you think x and y represent?

Lesson 6: Slope in Context: Interpreting Rates of Change

Slope as a rate of change with units — reading it off a graph, interpreting the intercept, average rate of change, and base-plus-commission models.

1. Reading a Rate of Change from a Graph

For a straight-line graph the slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

is the same no matter which two points on the line are used. When the axes measure real quantities, this constant ratio is the **rate of change**: how much the vertical quantity changes for each one-unit increase in the horizontal quantity. A positive slope means the vertical quantity is *increasing*; a negative slope means it is *decreasing*. The units come straight from the axes:

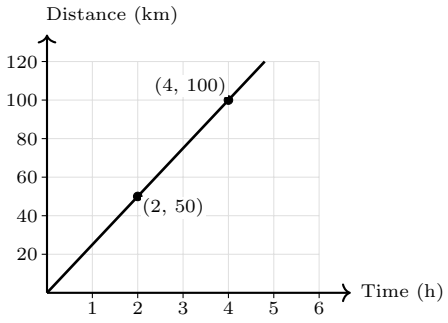
$$\text{vertical units per horizontal unit.}$$

A rate of change is therefore answered with a sentence carrying units — "the distance is increasing at 25 km/h" — not with a bare number.

Example 1 — Reading a rate of change off a graph

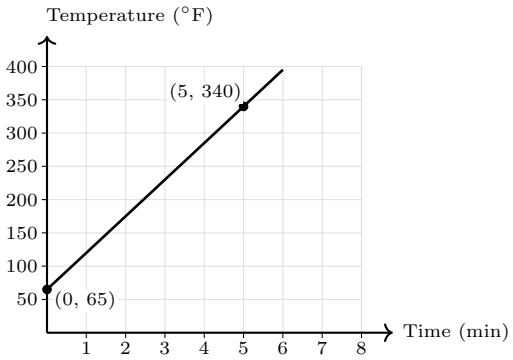
Each graph below shows a linear relationship between two real-world quantities.

(a) The distance travelled by a cyclist as a function of time.



Use $m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$ to calculate the slope of the line, then complete: "The distance is _____ (increasing/decreasing) at the rate of _____ km/h."

(b) The temperature inside an oven as a function of time.



Calculate the slope of the line. What units should be used for this rate of change? Then complete: "The temperature is (increasing/decreasing) at the rate of _____ _____ per _____."

2. Writing the Equation of a Rate-of-Change Graph

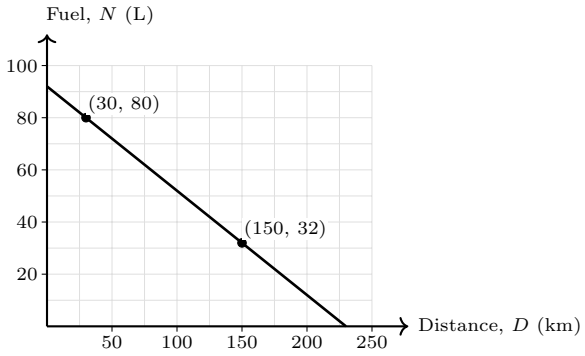
Given the rate of change (the slope) and the coordinates of one point on the line, substitute the point into $y = mx + b$ and solve for b, or use slope–point form, $y - y_1 = m(x - x_1)$, from Lesson 4 and rearrange. Either works whether the rate is positive or negative.

The vertical intercept b carries meaning too: it is the value of the vertical quantity when the horizontal quantity is 0 — a starting amount, a full tank, a fixed fee. State what it means in context, with units, just as you do for the slope.

Example 2 — Equation and interpretation from a fuel-consumption graph

---

The graph shows the fuel remaining in the tank of a delivery van, N litres, as a function of the distance driven, D kilometres.



(a) Calculate the slope of the line. What units apply here? Then complete: "The amount of fuel in the tank is (increasing/decreasing) at the rate of _____ _____ per _____."

(b) Determine the equation of the line in the form $N = mD + b$, and state what the value of b means for the van.

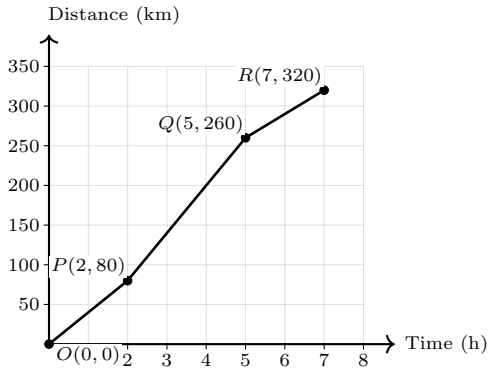
3. Average Rate of Change Between Two Points

Not every situation runs at one steady rate — a car may travel quickly on the highway for a while and slowly through towns the next while. Join any two data points with a straight segment: the slope of that segment is the **average rate of change** between the two points, the constant rate that would have produced the same overall change even though the actual rate varied in between. Different pairs of points on the same graph give different average rates of change.

Key idea. On a graph of distance as a function of time, the slope of the segment joining two points is the **average speed** between those two points.

Example 3 — Average speed over each leg of a trip

The graph records a road trip by car at four points: $O(0,0)$, $P(2,80)$, $Q(5,260)$ and $R(7,320)$, where distance is in kilometres and time is in hours. Calculate the average speed of the car over each interval below.



(a) between O and P

(b) between P and Q

(c) between Q and R

(d) for the whole trip, O to R

4. Modelling Base Pay Plus Commission

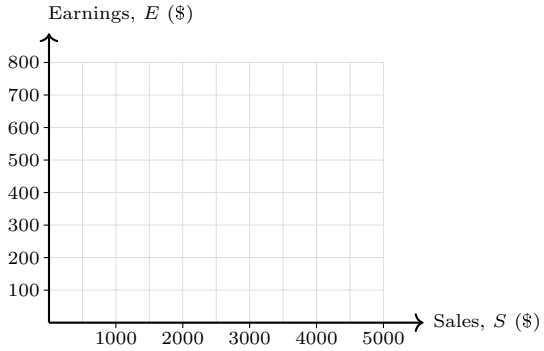
Many jobs pay a fixed base amount plus a variable amount that depends on performance, such as commission on sales. If earnings E depend on sales S linearly, $E = mS + b$, then the slope m is the **commission rate** (a decimal, usually quoted as a percent) and the vertical intercept b is the **fixed base pay** — the amount earned on zero sales.

Example 4 — Base salary plus commission

Maria is paid a weekly base salary plus commission for selling kitchen appliances. One week her sales totalled \$2 800 and she earned \$560. The next week her sales totalled \$4 600 and she earned \$740.

(a) On the grid, plot the two ordered pairs (Sales, Earnings) described above and join them.

(b) Calculate the slope of the line segment joining the two points, and explain what it represents in this context.



(c) State, as a percent, Maria's commission rate. Then calculate her fixed weekly base salary and write the equation of the line in the form $E = mS + b$.

(d) Three weeks ago Maria earned \$410. Use your equation from part (c) to calculate her sales that week.

Lesson 7: Equations of Linear Relations Unit Review

Pulling the unit together — reading a line from its equation, building its equation from a graph or a condition, and using parallel and perpendicular slopes.

1. Reading Slope and Intercepts From an Equation

An equation in general form $Ax + By + C = 0$ becomes slope–intercept form $y = mx + b$ once y is isolated: the coefficient of x is then the slope and the constant is the y-intercept. Either intercept can also be read straight from general form without solving for y — substitute $x = 0$ and solve for y to get the y-intercept, or $y = 0$ and solve for x to get the x-intercept.

Example 1 — Slope and intercepts read off an equation

(a) State the slope of the line with equation $5y = 4x - 20$.

(b) The y-intercept of the line with equation $y = -3x + 18$ is

A. 3 B. -3 C. 18 D. -18

(c) The point where the line $6x - 3y + 21 = 0$ crosses the y-axis is

A. $(0, 21)$ B. $(0, 7)$ C. $(0, -7)$ D. $(0, -21)$

(d) Which equation represents a line with slope -5 and y-intercept 7?

A. $y = 7x - 5$ B. $y = -\frac{1}{5}x + 7$

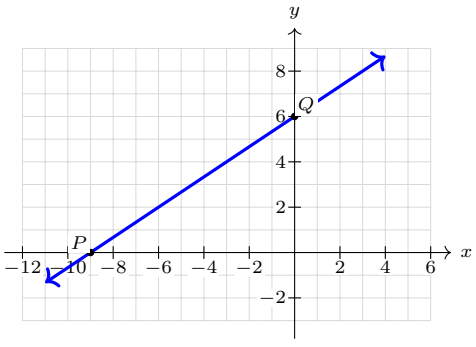
C. $y = -5x + 7$ D. $y = -5x - 7$

2. Writing the Equation of a Line From a Graph

A line drawn through lattice points supplies two exact coordinates. Compute the slope between them, then substitute one point into $y = mx + b$ or into $y - y_1 = m(x - x_1)$ and rearrange into whichever form the question asks for.

Example 2 — From a graph to general form

The diagram shows the line through P and Q, both of which have integer coordinates.



(a) Determine the equation of line PQ, in general form.

(b) Line PQ also passes through $(10.3, k)$. Determine k, to the nearest tenth.

3. Parallel and Perpendicular Lines

Two non-vertical lines are **parallel** exactly when their slopes are equal; all vertical lines are parallel. Two non-vertical lines are **perpendicular** exactly when their slopes are negative reciprocals of one another — equivalently, when the two slopes multiply to -1 (the turned slope triangle in Lesson 2, Section 3 shows why). A vertical line $x = k$ has an undefined slope, so it is excluded from that test; it is perpendicular to every horizontal line $y = k$, whose slope is 0.

Example 3 — Testing lines for perpendicularity

(a) Which of the following is the equation of a line perpendicular to $4y + x + 8 = 0$?

A. $y = 4x$ B. $y = x$ C. $y = \frac{1}{4}x$ D. $y = -\frac{1}{4}x$

(b) The equations of four lines are

$$1)\ 4x - y = 0 \qquad 2)\ 4x + y - 5 = 0 \qquad 3)\ x - 4y + 3 = 0 \qquad 4)\ x + 4y - 7 = 0.$$

Which pairs of lines are perpendicular?

A. 1) and 2) only B. 1) and 4) only C. both 1)&4) and 2)&3) D. both 1)&2) and 2)&3)

(c) Which of the following lines is/are perpendicular to the line $6x + y + 5 = 0$?

$$i)\ 6y + x = 3 \qquad ii)\ 6y - x = 3 \qquad iii)\ y = 6x + 3 \qquad iv)\ 6y = -x - 3$$

A. *i*) and *iii*) only B. *ii*) only C. *iv*) only D. some other combination

4. Applying Perpendicularity: Altitudes of a Triangle

The **altitude** from a vertex of a triangle is the segment drawn from that vertex perpendicular to the opposite side. Its equation is built like any other perpendicular: take the negative reciprocal of the side's slope, then substitute the vertex into slope–point form.

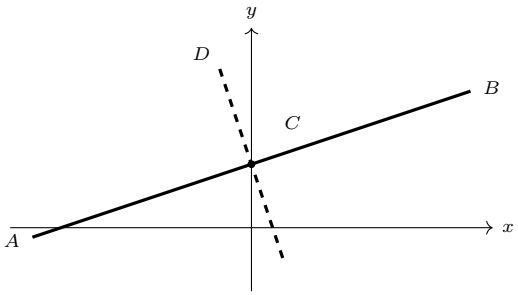
Example 4 — Altitudes and perpendiculars through an intercept

(a) In triangle DEF, side DE has slope $\frac{2}{5}$ and vertex F has coordinates $(-6, 9)$. The equation of the altitude from F to DE, written in the form $ax + by = c$, is

A. $2x - 5y = -57$ B. $5x + 2y = 33$ C. $5x + 2y = -12$ D. $5x + 2y = -33$

Note: $ax + by = c$, with the constant on the right side, is a different arrangement from general form $Ax + By + C = 0$. Some resources call $ax + by = c$ *standard form*. To change one to the other, move the constant across: $5x + 2y = -12$ is $5x + 2y + 12 = 0$ in general form.

(b) In the diagram, the equation of AB is $x - 3y + 6 = 0$. Line AB crosses the y-axis at C, and CD is drawn perpendicular to AB.



The equation of CD is

A. $x + 3y - 4 = 0$ B. $3x + y - 2 = 0$

C. $3x - y + 2 = 0$ D. $3x + y - 6 = 0$

5. Rates of Change in Applied Contexts

The slope of a linear model is a **rate of change**: the change in the output quantity per unit change in the input quantity. It carries the units of output per unit of input, and two data points are enough to calculate it even when no equation has been given.

Example 5 — A rate of change from two measurements

The temperature at the base of a hiking trail is 15.4 °C. At an elevation of 2 400 m above the base, the temperature is 1.0 °C. Determine the rate of temperature change, in °C per km, to the nearest tenth.